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(54) **INFRARED BAND PASS FILTER HAVING Si:NH LAYERS WITH REFRACTION INDEX GREATER THAN 3.5**

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See application file for complete search history.

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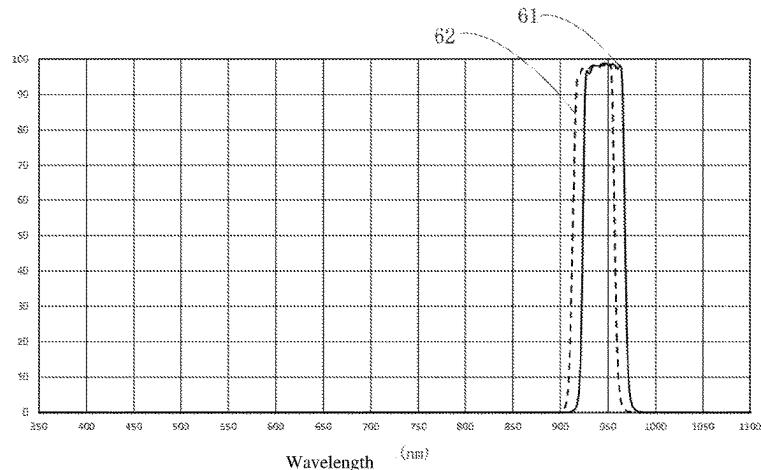
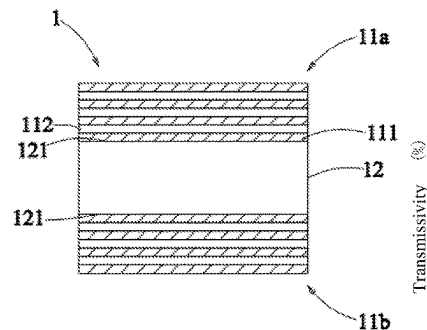
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(57) **ABSTRACT**

The present disclosure relates to an infrared band pass filter, which comprises a first multilayer film. The first multilayer film including a plurality of Si:NH layers and a low refraction index layer. The plurality of low refraction index layers are stacked with Si:NH layers alternatively; wherein the difference between the refraction index of Si:NH layer and the refraction index of the low refraction index layer is greater than 0.5. The infrared band pass filter has a pass band in a wavelength range of 800 nm and 1100 nm, and when the incident angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted less than 12 nm, and the infrared band pass filter of the present disclosure

(Continued)



can be used to enhance the 3D image resolution when applied to a 3D imaging system.

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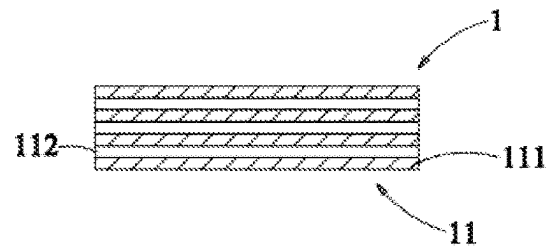


Fig. 1

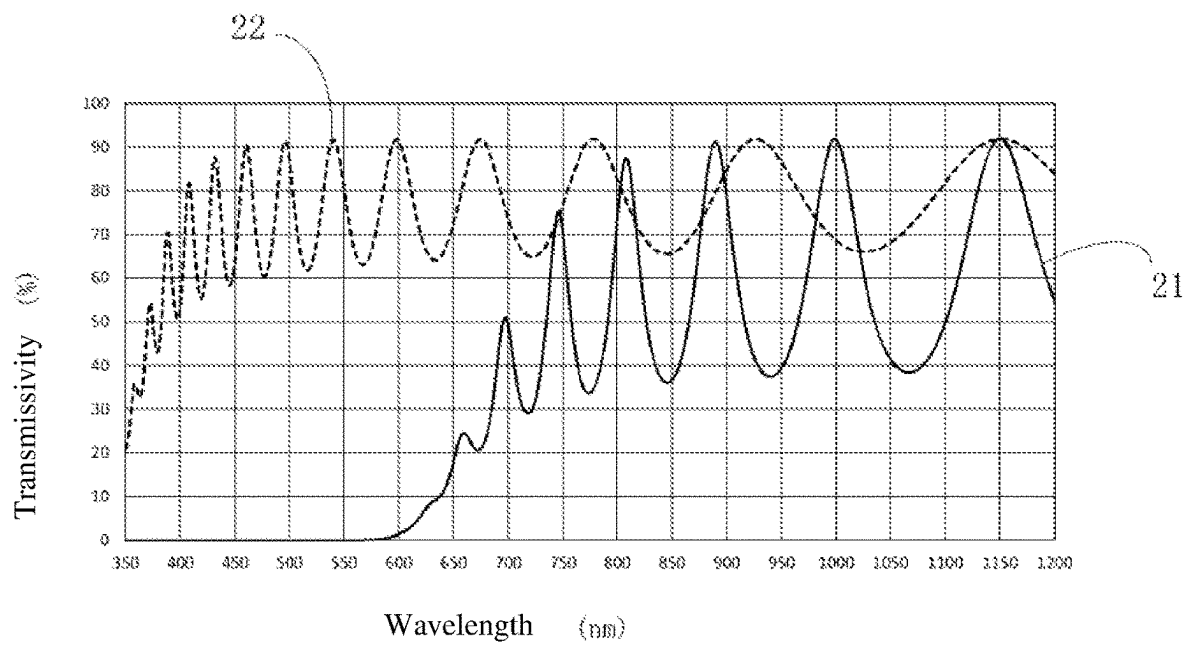


Fig. 2

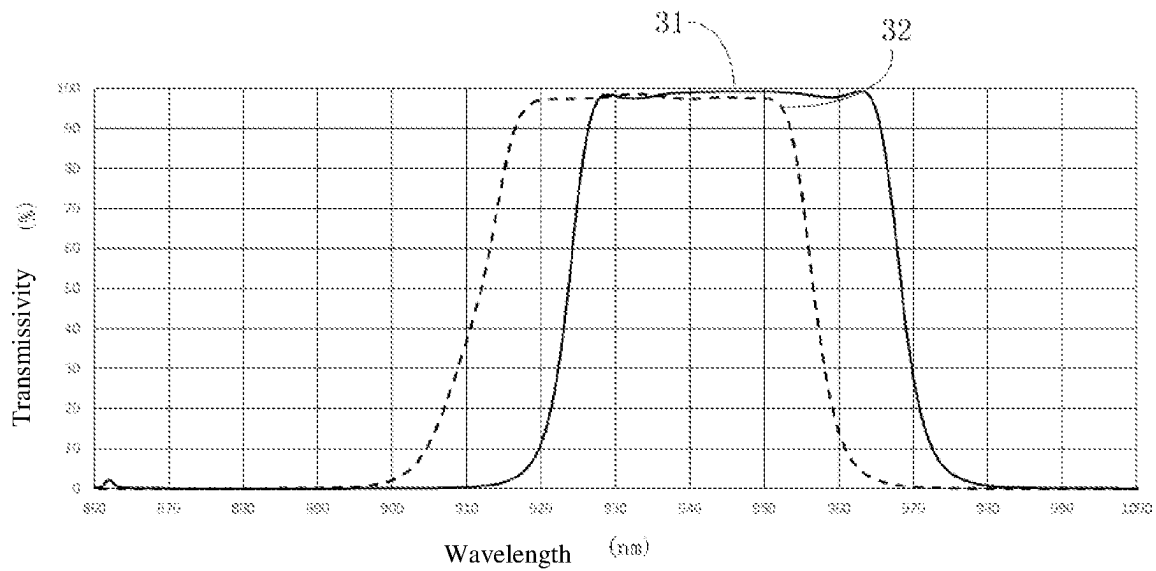


Fig. 3

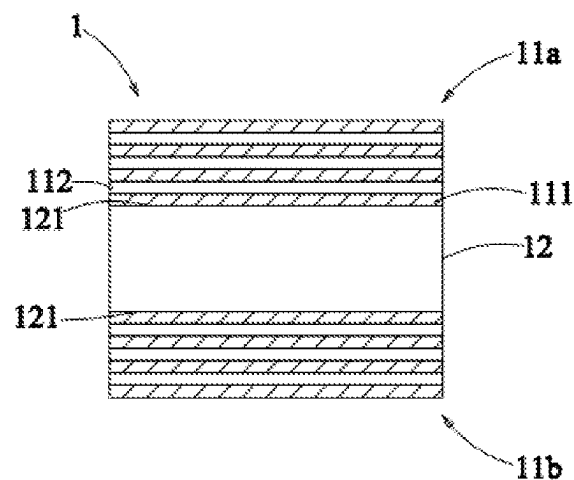


Fig. 4

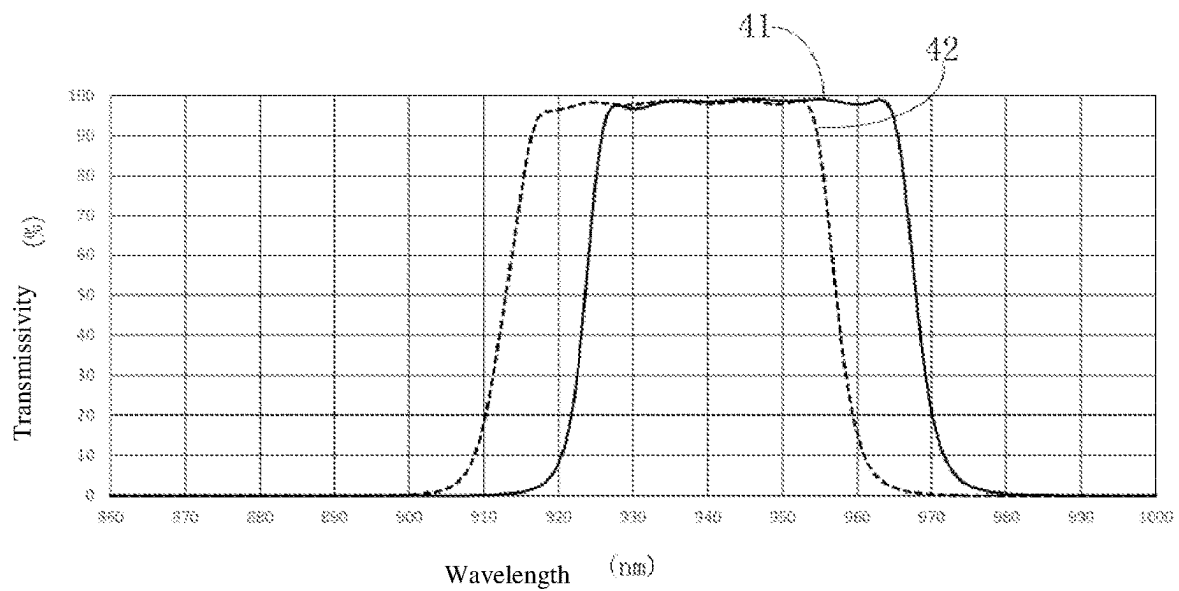


Fig. 5

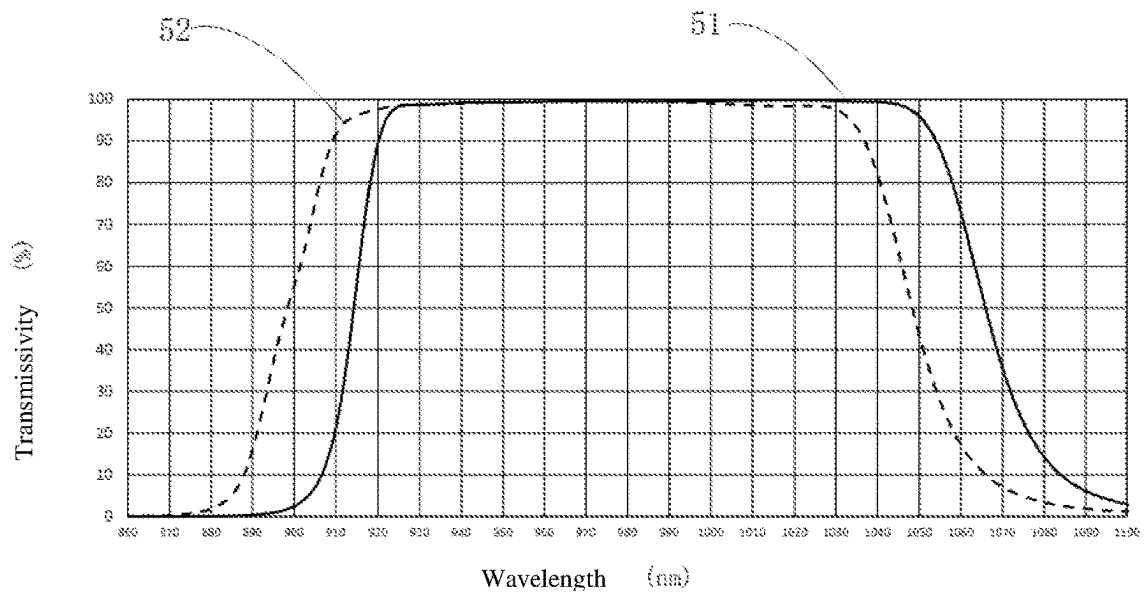


Fig. 6

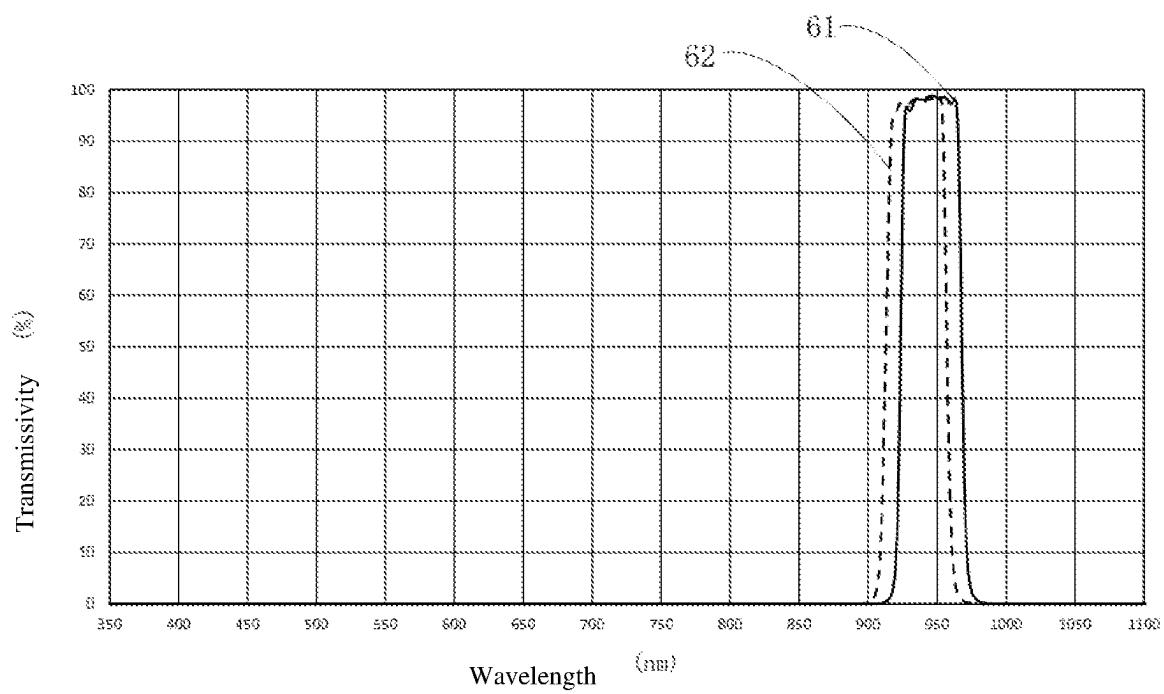


Fig. 7

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INFRARED BAND PASS FILTER HAVING Si:NH LAYERS WITH REFRACTION INDEX GREATER THAN 3.5

CROSS REFERENCE TO RELATED APPLICATION

This application is a divisional application of U.S. patent application Ser. No. 16/508,936, filed on Jul. 11, 2019, which claims the priority benefit of Chinese Patent Application Serial Number 201810921532.7, filed on Aug. 14, 2018, the full disclosure of which is incorporated herein by reference.

BACKGROUND

Technical Field

The present disclosure relates to the technical field of optical filtering, and more particularly to an infrared band pass filter.

Related Art

The existing infrared band pass filter comprises an optical substrate having a multilayer film formed on one surface of the optical substrate, and a multilayer anti-refraction layer formed on the other surface thereof. The multilayer film is formed by alternately depositing a plurality of high refractive index layers and a plurality of low refractive index layers. The material of the high refraction index layer is selected from the group consisting of Ti_3O_5 , Ta_2O_5 , ZrO_2 , Nb_2O_5 and TiO_2 and the material of the low refraction index layer is selected from one of the MgF_2 and SiO_2 .

The infrared band pass filter has a pass band in the wavelength range between 800 nm and 1100 nm. When the incident angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted between 31 nm and 34 nm. The center wavelength of the pass band is greatly shifted due to the angle changed, and when the infrared band pass filter is applied to the three-dimensional (3D) imaging system, the problem of unrecognizable or identification failure occurs when the light is received at a large angle.

SUMMARY

The present disclosure provides an infrared band pass filter, comprising: a first multilayer film, wherein the first multilayer film comprises: a plurality of Si:NH layers, wherein the refraction index of each Si:NH layer is larger than 3.5 in the wavelength range between about 800 nm and 1100 nm, and the extinction coefficient of the each Si:NH layer is smaller than 0.0002; and plurality of low refraction index layers a plurality of low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein the refraction index of each low refraction index layer is smaller than 3 in the wavelength range between 800 nm and 1100 nm; wherein an difference value between the refraction index of the each Si:NH layer and the low refraction index layer is larger than 0.5.

According to one embodiment of the present disclosure, the infrared band pass filter further comprises a substrate; the first multilayer film is formed on one surface of the substrate.

According to one embodiment of the present disclosure, the infrared band pass filter further comprises a second

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multilayer film which is formed on the other surface of the substrate, being opposite to the first multilayer film.

According to one embodiment of the present disclosure, the second multilayer film comprises: a plurality of Si:NH layers, wherein the refraction index of each Si:NH layer is larger than 3.5 in the wavelength range between about 800 nm and 1100 nm, and the extinction coefficient of the each Si:NH layer is smaller than 0.0002; and plurality of low refraction index layers a plurality of low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein the refraction index of each low refraction index layer is smaller than 3 in the wavelength range between 800 nm and 1100 nm; wherein an difference value between the refraction index of the each Si:NH layer and the low refraction index layer is larger than 0.5.

According to one embodiment of the present disclosure, the infrared band pass filter has a pass band partially overlapping with the wavelength range between 800 nm and 1100 nm, and the center wavelength of pass band is within the wavelength range between 800 nm and 1100 nm, and having a blocking level greater than OD5 in the wavelength range between 400 nm and 1100 nm.

According to one embodiment of the present disclosure, the infrared band pass filter has a pass band partially overlapping with the wavelength range between 800 nm and 1100 nm, and the center wavelength of the pass band is in the wavelength range between 800 nm and 1100 nm.

According to one embodiment of the present disclosure, when the incident angle is changed from 0 degrees to 30 degrees, and the center wavelength of the pass band is shifted less than 12 nm.

According to one embodiment of the present disclosure, the sum of the number of layers of the Si:NH layer and the number of layers of the low refraction index layer is less than 39 layers.

According to one embodiment of the present disclosure, the total thickness of the plurality of Si:NH layers and the plurality of low refraction index layers is less than 4.5 μm .

According to one embodiment of the present disclosure, the hardness of infrared band pass filter is greater than Mohs hardness 7.

According to one embodiment of the present disclosure, the material of the low refraction index layer is selected from the group consisting of SiO_2 , Al_2O_3 , TiO_2 , Nb_2O_5 , Ta_2O_5 , $SiOH$ and $SiNOH$.

According to one embodiment of the present disclosure, the material of the low refraction index layer is SiO_2 , the refraction index of the low refraction index layer is between 1.4 and 1.5 in a wavelength range between 800 nm and 1100 nm.

According to the embodiment of the present disclosure, when the incident angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted less than 11.8 nm.

According to one embodiment of the present disclosure, the material of the low refraction index layer is $SiNOH$, the refraction index of the low refraction index layer is between 1.5 and 2 in a wavelength range between 800 nm and 1100 nm.

According to one embodiment of the present disclosure, when the incident angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted less than 11.1 nm.

According to one embodiment of the present disclosure, the hardness of infrared band pass filter is greater than Mohs hardness 9.

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According to one embodiment of the present disclosure, the extinction coefficient of the Si:NH layer is less than 0.00012 in the wavelength range between 800 nm and 1100 nm.

According to one embodiment of the present disclosure, the transmissivity of pass band is greater than 90% in the wavelength between 800 nm and 1100 nm.

Compared with the prior art, the present disclosure can obtain the following technical effects:

The present disclosure provides an infrared band pass filter. The infrared band pass filter of the present disclosure is formed by alternatively stacking a Si:NH layer and a low refraction index layer, which has a pass band in a wavelength range between 800 nm and 1100 nm. When the incident angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted less than 12 nm, and the infrared band pass filter of the present disclosure can be applied to the 3D imaging system to improve the 3D image analysis capability.

It should be understood, however, that this summary may not contain all aspects and embodiments of the present disclosure, that this summary is not meant to be limiting or restrictive in any manner, and that the disclosure as disclosed herein will be understood by one of ordinary skill in the art to encompass obvious improvements and modifications thereto.

BRIEF DESCRIPTION OF THE DRAWINGS

The features of the exemplary embodiments believed to be novel and the elements and/or the steps characteristic of the exemplary embodiments are set forth with particularity in the appended claims. The Figures are for illustration purposes only and are not drawn to scale. The exemplary embodiments, both as to organization and method of operation, may best be understood by reference to the detailed description which follows taken in conjunction with the accompanying drawings in which:

FIG. 1 shows the schematic diagram of an infrared band pass filter according to a first embodiment of present disclosure;

FIG. 2 shows a curve graph of the transmission spectrum of the Si:NH layer according to the first embodiment of the present disclosure;

FIG. 3 shows curve graph of the transmission spectrum of incident angle from 0 degrees to 30 degrees of the infrared band pass filter;

FIG. 4 shows the infrared band pass filter according to a second embodiment;

FIG. 5 shows the curve graph of the transmission spectrum of incident angle from 0 degrees to 30 degrees of the first multilayer film according to the second embodiment;

FIG. 6 shows the curve graph of the transmission spectrum of incident angle from 0 degrees to 30 degrees of the second multilayer film according to the second embodiment; and

FIG. 7 shows the curve graph of the transmission spectrum of incident angle from 0 degrees to 30 degrees of the infrared band pass filter according to the second embodiment.

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DETAILED DESCRIPTION OF THE EMBODIMENTS

The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the disclosure are shown. This present disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this present disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art.

Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but function. In the following description and in the claims, the terms “include/including” and “comprise/comprising” are used in an open-ended fashion, and thus should be interpreted as “including but not limited to”. “Substantial/substantially” means, within an acceptable error range, the person skilled in the art may solve the technical problem in a certain error range to achieve the basic technical effect. The following description is of the best-contemplated mode of carrying out the disclosure. This description is made for the purpose of illustration of the general principles of the disclosure and should not be taken in a limiting sense. The scope of the disclosure is best determined by reference to the appended claims.

Moreover, the terms “include”, “contain”, and any variation thereof are intended to cover a non-exclusive inclusion. Therefore, a process, method, object, or device that includes a series of elements not only includes these elements, but also includes other elements not specified expressly, or may include inherent elements of the process, method, object, or device. If no more limitations are made, an element limited by “include a/an . . .” does not exclude other same elements existing in the process, the method, the article, or the device which includes the element.

In the following embodiment, the same reference numerals is used to refer to the same or similar elements throughout the disclosure.

FIG. 1 shows the schematic diagram of an infrared band pass filter according to a first embodiment of present disclosure. As shown in FIG. 1, the embodiment provides an infrared band pass filter 1 adapted for a 3D imaging system. The infrared band pass filter 1 comprises a first multilayer film 11. The first multilayer film 11 comprises a plurality of Si:NH layers 111 and a plurality of low refraction index layers 112, and the plurality of low refraction index layers 112 are stacking with the plurality of Si:NH layers alternatively. The refraction index of each Si:NH layer 111 is larger than 3.5 in the wavelength range between 800 nm and 1100 nm, and the extinction coefficient of the each Si:NH layer is smaller than 0.0002. In another embodiment, the extinction coefficient of the each Si:NH layer less than 0.00012. The refraction index of each low refraction index layer 112 is smaller than 3 in the wavelength range between 800 nm and 1100 nm; wherein a difference value between the refraction index of the each Si:NH layer 111 and the low refraction index layer 112 is larger than 0.5.

In the embodiment, the sum of the number of Si:NH layers 111 and the number of low refraction index layer 112 is smaller than 39 layers. The total thickness of the plurality of Si:NH layers 111 and the plurality of low refraction index layers 112 is smaller than 4.5 μm . In the embodiment, the

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infrared band pass filter **1** has some band pass overlapping the wavelength range between 800 nm and 1100 nm. The center wavelength of the band pass is located in the wavelength range between 800 nm and 1100 nm.

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Ta₂O₅, the silicon target of the sputtering machine is replaced with an aluminum target, a titanium target, a tantalum target or a tantalum target, and formed by the above method.

TABLE 1

Material	TG4 gas	ICP gas			At 940 nm n	At 940 nm k
	Argon	Argon	oxygen	Ammonia	(Opt)	(Opt)
SiNOH	60	500	5	30	3.42022	8.35E-05
SiNOH	60	500	10	30	3.267666	6.24E-05
SiNOH	60	500	30	30	2.781278	3.23E-05
SiNOH	60	500	70	30	2.176382	2.27E-04
SiNOH	60	500	76	30	2.125067	2.60E-04
SiNOH	60	500	80	30	2.037629	1.85E-04
SiNOH	60	500	88	30	1.986488	1.94E-04
SiNOH	60	500	95	30	1.908388	1.42E-04
SiNOH	60	500	104	30	1.846262	1.69E-04
SiNOH	60	500	125	30	1.576734	1.26E-05

While the incidence angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted less than 12 nm. Transmissivity of the band pass is larger 90% in the wavelength range between 800 nm and 1100 nm. In the embodiment, the hardness of the infrared band pass filter is larger than Mohs' hardness 7. In the embodiment, the center wavelength of the big angle of the infrared band pass filter **1** being smaller can solve the 3D imaging analyze problem effectively, while the thickness of the infrared band pass filter **1** of the present embodiment is slimmer than the infrared band pass filter of the prior technology so as to decrease the warping phenomenon of the infrared band pass filter **1**.

In the embodiment, manufacture method of the infrared band pass filter **1** is to form the Si:NH layer **111** and the low refraction index layer **112** by the sputter deposition. The process of the sputter deposition is that the mid frequency sputter power source output 50 KHz voltage through the inside electrical network, respectively applied on a pair of silicon target on the sputtering machine. Argon is inputted into the silicon target via the air duct so as to generate light and discharge and generate the steady plasma.

Due to the reaction chamber of the sputtering machine has the quadrature field being vertical to the electrical filed, the movement direction of the electron is changed and limited in the certain space so as to increase the collision probability with the argon and ionization effect of electron.

The argon ion elected from ionization effect crash the silicon target quickly in the electrical field, and the mass silicon atom is sputtered out, while inductively coupled plasma (ICP) is mixed with ammonia to generate the Si:NH layer **111** which has the refraction index larger than 3.5 in the wavelength range between 800 nm and 1100 nm. The low refraction index layer **112** is selected from the group consisting of SiO₂、SiOH or SiNOH, and the ICP is mixed with oxygen/ammonia to generate the low refraction index layer which the refraction index is smaller than 3 in the wavelength range between 800 nm and 1000 nm. If the low refraction index layer **112** is Al₂O₃、TiO₂、Nb₂O₅ or

As shown in Table 1, the refraction index of the low refraction layer **112** of the embodiment will changed along with the change of the oxygen flow via the spurting process in the wavelength 940 nm. When the oxygen flow is becoming bigger, then the refraction index of the low refraction index layer **112** in the wavelength 940 nm is becoming lower. When the low refraction index layer **112** is SiNOH, the refraction index of the low refraction index layer **112** is between 1.5 and 2 in the wavelength range between 800 nm and 1100 nm. When the low refraction index layer **112** is SiO₂, the refraction index of the low refraction index layer **112** is between 1.4 and 1.5 in the wavelength range between 800 nm and 1100 nm.

FIG. 2 shows a curve graph of the transmission spectrum of the Si:NH layer according to the first embodiment of the present disclosure. As shown in FIG. 2, the curve **21** is the transmission spectrum of the Si:NH layer and the curve **22** is the transmission spectrum of the Ti₃O₅ layer. The absorption effect of the Si:NH layer **111** is better than the Ti₃O₅ layer used by the infrared band pass filter in the wavelength range between 350 nm and 700 nm. The infrared band pass filter **1** having Si:NH layer is applied in the 3D image system so as to increase the image analyze function of the 3D image system.

The material of the low refraction index layer **112** is selected from the group consisting of SiO₂、Al₂O₃、TiO₂、Nb₂O₅、Ta₂O₅、SiOH and SiNOH. The material of the low refraction index layer **112** is SiO₂, the refraction index of the low refraction index layer **112** is between 1.4 and 1.5 in the wavelength range between 800 nm and 1100 nm. The sum of the number of Si:NH layers **111** and the number of low refraction index layer **112** is 39 layers. The sum of thickness of the plurality of Si:NH layers **111** and the plurality of low refraction index layers **112** is 4.4 um. Table 2 shows the thickness list of each Si:NH layer **111** and each low refraction index layer **112**.

TABLE 2

Number of layers	material	thickness (nm)	Number of layers	material	thickness (nm)	Number of layers	material	Thickness (nm)
1	SiO ₂	82.19	14	Si:NH	21.84	27	SiO ₂	121.07
2	Si:NH	253.51	15	SiO ₂	124.75	28	Si:NH	80.92
3	SiO ₂	118.19	16	Si:NH	99.93	29	SiO ₂	300.84
4	Si:NH	41.54	17	SiO ₂	99.75	30	Si:NH	94.95
5	SiO ₂	306.22	18	Si:NH	41.78	31	SiO ₂	102.9
6	Si:NH	92.21	19	SiO ₂	295.9	32	Si:NH	49.42
7	SiO ₂	78.43	20	Si:NH	61.93	33	SiO ₂	113.78
8	Si:NH	69.85	21	SiO ₂	108.78	34	Si:NH	15.21
9	SiO ₂	70.53	22	Si:NH	31.22	35	SiO ₂	288.75
10	Si:NH	57.16	23	SiO ₂	106.84	36	Si:NH	4.87
11	SiO ₂	280.38	24	Si:NH	82.33	37	SiO ₂	89.27
12	Si:NH	233.19	25	SiO ₂	128.68	38	Si:NH	17.57
13	SiO ₂	103.92	26	Si:NH	132.25	39	SiO ₂	15.54

FIG. 3 shows curve graph of the transmission spectrum of incident angle from 0 degrees to 30 degrees of the infrared band pass filter. As shown in FIG. 3, it shows the transmission spectrum curve **31** in the incident angle of 0 degrees and the transmission spectrum curve **32** in the incident angle of 30 degrees. The center wavelength of the band pass is 946 nm in the incident angle of 0 degrees in the embodiment. The center wavelength of the band pass is 934.2 nm in the incident angle of 30 degrees in the embodiment. Therefore, when the incident angle of the infrared band pass filter is changed from 0 degrees to 30 degrees, the offset amplitude of the center wavelength of the band pass is smaller than 11.8 nm. The transmissivity of the band pass is larger than 90% in the wavelength range between 800 nm and 1100 nm.

FIG. 4 shows the infrared band pass filter according to a second embodiment. As shown in FIG. 4, the infrared band pass filter **1** comprises a first multilayer film **11a**, a second multilayer film **11b** and an optical substrate **12**. The first multilayer film **11a** and the second multilayer film **11b** are respectively arranged the two opposite surfaces **121** of the optical substrate **12**. Specifically, the first multilayer film **11a** and the second multilayer film **11b** respectively has a structure of the plurality of Si:NH layers **111** alternatively stacking with the plurality of low refraction index layers **112**. The manufacture methods of the Si:NH layer **111** and the low refraction index layer **112** are described above. In the embodiment, the thickness of the infrared band pass filter is slimmer than the infrared band pass filter of the prior technology, and while the infrared band pass filter **1** of the embodiment has two multilayer films, the two multilayer films will be maintained balance via film stress so as to decrease the warp phoneme happen of the infrared band pass filter.

In the infrared band pass filter **1** of the embodiment, the material of the low refraction index layer **112** of the first multilayer film **11a** is SiNOH. The sum of the number of the Si:NH layer **111** and the number of the low refraction index layer **112** is 39 layers. The total thickness of the plurality of Si:NH layer **111** and the plurality of low refraction index layer **112** is 4.3 um. Table 3 shows the thickness list of the each Si:NH layer **111** and each low refraction index layer **112**.

TABLE 3

Number of layers	Material	Thickness (nm)
1	SiNOH	297
2	Si:NH	15.1

TABLE 3-continued

Number of layers	Material	Thickness (nm)
3	SiNOH	38.88
4	Si:NH	115.6
5	SiNOH	211.44
6	Si:NH	280.91
7	SiNOH	62.06
8	Si:NH	79.67
9	SiNOH	50.49
10	Si:NH	70.88
11	SiNOH	81.85
12	Si:NH	306.07
13	SiNOH	19.97
14	Si:NH	104
15	SiNOH	103.41
16	Si:NH	124.59
17	SiNOH	38.43
18	Si:NH	99.3
19	SiNOH	55.11
20	Si:NH	295.27
21	SiNOH	28.61
22	Si:NH	108.38
23	SiNOH	74.2
24	Si:NH	105.09
25	SiNOH	114.44
26	Si:NH	127.7
27	SiNOH	69.68
28	Si:NH	121.36
29	SiNOH	6.85
30	Si:NH	288.78
31	SiNOH	44.21
32	Si:NH	102.59
33	SiNOH	14.23
34	Si:NH	112.35
35	SiNOH	83.11
36	Si:NH	299.68
37	SiNOH	14.19
38	Si:NH	89.14
39	SiNOH	80.45

Please refer to FIG. 5, which shows the curve graph of the transmission spectrum of incident angle from 0 degrees to 30 degrees of the first multilayer film according to the second embodiment of the present disclosure; as shown in FIG. 5, which shows the transmission spectrum curve **41** at the incident angle 0 degrees and the transmission spectrum curve **42** at the incident angle 30 degrees. In the present embodiment, the center wavelength of band pass is in 939.9 nm at an incident angle of 0 degrees and the center wavelength is in 932.8 nm at an incident angle of 30 degrees. Therefore, when the infrared band pass filter of the present embodiment is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted less than 11.1 nm. The transmission of the pass band in the wavelength range between 800 nm and 1100 nm is more than 90%.

However, the material of the low refraction index layer **112** of the second multilayer film **11b** of the present embodiment also uses SiNOH, and the total number of layers of the Si:NH layer **111** and the low refraction index layer **112** is 32 layers. The total thickness of the plurality of Si:NH layers **111** and a plurality of low refraction index layers **112** of the second multilayer film **11b** is 2.7 μm . The Table 3 below lists the thicknesses of each Si:NH layer **111** and the low refraction index layer **112**.

TABLE 3

Number of layers	Material	Thickness (nm)
1	SINOH	20.84
2	SINH	85.77
3	SINOH	121.13
4	SINH	72.85
5	SINOH	128.99
6	SINH	138
7	SINOH	108.69
8	SINH	49.76
9	SINOH	150.76
10	SINH	110.88
11	SINOH	47.85
12	SINH	121.71
13	SINOH	145.74
14	SINH	127.98
15	SINOH	63.26
16	SINH	122.03
17	SINOH	31.33
18	SINH	91.1
19	SINOH	46.32
20	SINH	101.34
21	SINOH	53.27
22	SINH	99.56
23	SINOH	57.74
24	SINH	71.18
25	SINOH	45.12
26	SINH	96.14
27	SINOH	52.1
28	SINH	91.74
29	SINOH	56.67
30	SINH	73.42
31	SINOH	19.03
32	SINH	72.95

Referring to FIG. 6 and FIG. 7 again, it shows the curve graph of the transmission spectrum of the second multilayer film of the second embodiment of the present disclosure at an incident angle of 0 degrees and 30 degrees, and the curve graph of the transmission spectrum of the infrared band pass filter of the second embodiment of the present disclosure at an incident angle of 0 degrees and 30 degrees. As shown in FIG. 6, the transmission spectrum curve **51** at the incident angle 0 degrees and the transmission spectrum curve **52** at the incidence angle 30 degrees. The second multilayer film **11b** of the present embodiment is used to increase the transmissivity of the infrared band pass filter. As can be seen from FIG. 6, the transmissivity of the pass band of the infrared band pass filter in the wavelength range between 800 nm and 1100 nm is greater than 90%, meanwhile, the outside of the pass band of the infrared band pass filter has a blocking level which is larger than OD5 in the wavelength range between 400 nm and 1100 nm. FIG. 7 shows the transmission spectrum curve **61** at an incident angle 0 degrees and the transmission spectrum curve **62** at an incident angle 30 degrees. The infrared band pass filter of the present embodiment, at an incident angle 0 degrees, has the center wavelength of the pass band being at 943.9 nm. The infrared band pass filter of this embodiment, at an incident angle 30 degrees, the center wavelength of pass band being

at 932.8 nm. Therefore, when the infrared band pass filter of the present embodiment is changed from 0 degrees to 30 degrees, the shift wavelength of the center wavelength of the pass band is less than 11.1 nm.

The Mohs hardness of the infrared band pass filter of the embodiment can achieve to 9 and 9.5. The Vickers hardness is about 2200, and the micro-hardness is 32630 MPa, thus demonstrating that the infrared band pass filter of the embodiment has good chemical resistance, and having a good chemical corrosion resistance under inorganic acid, less than 30% caustic soda solution or organic acid, also being a high performance electrical insulation material. The material of the low refraction index layer **112** of the first multilayer film **11a** and the second multilayer film **11b** of the present embodiment can also be made of other materials such as SiO_2 、 Al_2O_3 、 TiO_2 、 Nb_2O_5 、 Ta_2O_5 or SiOH. The stacked of the Si:NH layer **111** and the low refraction index layer **112** of the first multilayer film **11a** and the stacked of the Si:NH layer **111** and the low refraction index layer **112** of the second multilayer film **11b** of the present embodiment are mutually change.

In summary, in one or more embodiments of the present disclosure, the present disclosure provides an infrared band pass filter, which having at least one multilayer film. The multilayer film is formed by the Si:NH layer and the low refraction index layer alternatively stacked, which having a pass band in a wavelength range between 800 nm and 1100 nm. When the incident angle is changed from 0 degrees to 30 degrees, the center wavelength of the pass band is shifted by less than 12 nm. The shift of the large-angle center wavelength of the infrared band pass filter of the present disclosure is small, and the infrared band pass filter of the present disclosure can improve the three-dimensional image resolution capability when applied to a three-dimensional imaging system.

It is to be understood that the term “comprises”, “comprising”, or any other variants thereof, is intended to encompass a non-exclusive inclusion, such that a process, method, article, or device of a series of elements not only includes those elements but also includes other elements that are not explicitly listed, or elements that are inherent to such a process, method, article, or device. An element defined by the phrase “comprising a . . .” does not exclude the presence of the same element in the process, method, article, or device that comprises the element.

Although the present disclosure has been explained in relation to its preferred embodiment, it does not intend to limit the present disclosure. It will be apparent to those skilled in the art having regard to this present disclosure that other modifications of the exemplary embodiments beyond those embodiments specifically described here may be made without departing from the spirit of the disclosure. Accordingly, such modifications are considered within the scope of the disclosure as limited solely by the appended claims.

What is claimed is:

1. An infrared band pass filter, comprising:

a first multilayer film, wherein the multilayer film comprises:

a plurality of Si:NH layers, wherein a refraction index of each Si:NH layer is larger than 3.5 in a wavelength range between 800 nm and 1100 nm; and

a plurality of first low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein a refraction index of each first low refraction index layer is smaller than 3 in the wavelength range between 800 nm and 1100 nm;

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a substrate, wherein the substrate is formed on the first multilayer film; and

a second multilayer film, formed on the other surface of the substrate, opposite to the first multilayer film, and wherein the second multilayer film comprises:

- a plurality of Si:NH layers, wherein a refraction index of each Si:NH layer is greater than 3.5 in the wavelength range between 800 nm and 1100 nm; and
- a plurality of second low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein a refraction index of each second low refraction index layer is less than 3 in the wavelength range between 800 nm and 1100 nm;

wherein the substrate directly contacts the Si:NH layer of the first multilayer film and the Si:NH layer of the second multilayer film.

2. The infrared band pass filter of claim 1, wherein a material of the first low refraction index layer is selected from Al₂O₃ and SiNOH, and a material of the second low refraction index layer is selected from Al₂O₃ and SiNOH.

3. The infrared band pass filter of claim 1, wherein a total thickness of the plurality of Si:NH layers in the first multilayer film is the same as a total thickness of the plurality of Si:NH layers in the second multilayer film, and a total thickness of the plurality of first low refraction index layers in the first multilayer film is the same as a total thickness of the plurality of second low refraction index layers in the second multilayer film.

4. The infrared band pass filter of claim 1, wherein a wavelength range in which a transmissivity of the first multilayer film is greater than 90% is smaller than a wavelength range in which a transmissivity of the second multilayer film is greater than 90%.

5. The infrared band pass filter of claim 1, wherein when an incident angle is changed from 0 degrees to 30 degrees, a center wavelength of a pass band of the infrared band pass filter is shifted less than 12 nm.

6. The infrared band pass filter of claim 1, wherein a layer farthest from the substrate in the first multilayer film is the first low refraction index layer, and a layer farthest from the substrate in the second multilayer film is the second low refraction index layer.

7. The infrared band pass filter of claim 6, wherein a material of the layer farthest from the substrate in the first multilayer film is SiNOH, and a material of the layer farthest from the substrate in the second multilayer film is SiNOH.

8. An infrared band pass filter, comprising:

- a first multilayer film, wherein the multilayer film comprises:
 - a plurality of Si:NH layers, wherein a refraction index of each Si:NH layer is larger than 3.5 in a wavelength range between 800 nm and 1100 nm; and
 - a plurality of first low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein a material of the first low refraction index layer is selected from the group consisting of SiO₂, Al₂O₃, TiO₂, Nb₂O₅, Ta₂O₅, SiOH and SiNOH;
- a substrate, wherein the substrate is formed on the first multilayer film; and
- a second multilayer film, formed on the other surface of the substrate, opposite to the first multilayer film, and wherein the second multilayer film comprises:
 - a plurality of Si:NH layers, wherein a refraction index of each Si:NH layer is greater than 3.5 in the wavelength range 800 nm and 1100 nm; and
 - a plurality of second low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein

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a material of the second low refraction index layer is selected from the group consisting of SiO₂, Al₂O₃, TiO₂, Nb₂O₅, Ta₂O₅, SiOH and SiNOH;

wherein the substrate directly contacts the Si:NH layer of the first multilayer film and the Si:NH layer of the second multilayer film.

9. The infrared band pass filter of claim 8, wherein a total thickness of the plurality of Si:NH layers in the first multilayer film is the same as a total thickness of the plurality of Si:NH layers in the second multilayer film, and a total thickness of the plurality of first low refraction index layers in the first multilayer film is the same as a total thickness of the plurality of second low refraction index layers in the second multilayer film.

10. The infrared band pass filter of claim 8, wherein a layer farthest from the substrate in the first multilayer film is the first low refraction index layer, and a layer farthest from the substrate in the second multilayer film is the second low refraction index layer.

11. The infrared band pass filter of claim 8, wherein a hardness of the infrared band pass filter is greater than Mohs hardness 7.

12. The infrared band pass filter of claim 8, wherein when an incident angle is changed from 0 degrees to 30 degrees, a shift amount of a center wavelength of a pass band of the infrared band pass filter is from 11.1 nm to 12 nm.

13. The infrared band pass filter of claim 8, wherein when an incident angle is changed from 0 degrees to 30 degrees, a center wavelength of a pass band of the infrared band pass filter is shifted less than 12 nm.

14. The infrared band pass filter of claim 8, wherein a difference value between the refraction index of the Si:NH layer and the refraction index of the first low refraction index layer in the first multilayer film is larger than 0.5, and a difference value between the refraction index of the Si:NH layer and the refraction index of the second low refraction index layer in the second multilayer film is larger than 0.5.

15. An infrared band pass filter, comprising:

- a first multilayer film, wherein the multilayer film comprises:
 - a plurality of Si:NH layers, wherein a refraction index of each Si:NH layer is larger than 3.5 in a wavelength range between 800 nm and 1100 nm; and
 - a plurality of first low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein a refraction index of each first low refraction index layer is smaller than 3 in the wavelength range between 800 nm and 1100 nm;
- a substrate, wherein the substrate is formed on the first multilayer film; and
- a second multilayer film, formed on the other surface of the substrate, opposite to the first multilayer film, and wherein the second multilayer film comprises:
 - a plurality of Si:NH layers, wherein a refraction index of each Si:NH layer is greater than 3.5 in the wavelength range between 800 nm and 1100 nm; and
 - a plurality of second low refraction index layers, stacking with the plurality of Si:NH layers alternatively, wherein a refraction index of each second low refraction index layer is less than 3 in the wavelength range between 800 nm and 1100 nm;

wherein the first multilayer film and the second multilayer film are symmetrical to each other with the substrate as a center plane of symmetry;

wherein a layer farthest from the substrate in the first multilayer film is the first low refraction index layer,

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and a layer farthest from the substrate in the second multilayer film is the second low refraction index layer.

16. The infrared band pass filter of claim 15, wherein a material of the first low refraction index layer is selected from Al₂O₃ and SiNOH, and a material of the second low 5 refraction index layer is selected from Al₂O₃ and SiNOH.

17. The infrared band pass filter of claim 15, wherein a material of the layer farthest from the substrate in the first multilayer film is SiNOH, and a material of the layer farthest from the substrate in the second multilayer film is SiNOH. 10

18. The infrared band pass filter of claim 15, wherein when an incident angle is changed from 0 degrees to 30 degrees, a shift amount of a center wavelength of a pass band of the infrared band pass filter is equal to or less than 11.1 nm. 15

19. The infrared band pass filter of claim 15, wherein when an incident angle is changed from 0 degrees to 30 degrees, a center wavelength of a pass band of the infrared band pass filter is shifted less than 12 nm. 20

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